

A.1 Additional Implementation Details

A.2 Qualitative Analysis of Video Caption Ablation

[illegible]

A.3 Qualitative Analysis of Intent-Aware Mechanism

In this section, we show more cases about intent-aware mechanism analysis. The proper intention guidance enables the framework to anticipate more accurate and correct future actions instead of general ones. For instance, intent-aware LLM predicts “pen” instead of “wood” in case 1 (Fig. 6), “flower” rather than “plant” in case 2 (Fig. 7), and “pants” instead of “cloth” in case 3 (Fig. 8).

Fig. 8: Intent-aware mechanism analysis (Case 3). Blue represents the inferred intent, green indicates the correctly anticipated verbs/nouns, and red is the main discrepancies in noun anticipation with and without intent-aware.

In this section, we present a diversity analysis of the five generated candidate sequences. As shown in Fig. 9, the proposed framework is capable not only of generating different future behaviors based on similar trajectories (e.g., “crafting with clay and sand” vs. “crafting with clay and sand using molds and tools”), but also of deducing multiple distinct future behaviors (e.g., “prepare and clean up after a meal” vs. “prepare and serve coffee with milk”).

Fig. 9: Diversity analysis of 5 generated candidate sequences. Blue represents the inferred intent, and green indicates the correctly anticipated verbs/nouns.

A.5 Additional Discussion

Recent LTA methods have witnessed significant progress of integrating LLMs and intention reasoning. Methods like INSIGHT [4] have demonstrated the value of explicit cognitive reasoning through hand-object features and reinforcement learning-based intention inference. However, these approaches share a fundamental limitation: cascaded multi-stage pipelines that separate action recognition from temporal reasoning. This design inherently suffers from error propagation—recognition errors from the first stage corrupt subsequent reasoning, particularly for rare actions and out-of-distribution scenarios.

Moreover, the use of online reinforcement learning (e.g., INSIGHT [4]) for intention inference introduces additional challenges. The training process requires careful hyperparameter tuning which raise concerns about sample efficiency and training stability. In addition, the reinforcement learning objective may encourage the model to exploit spurious correlations in the training data, leading to overfitting problem. The separation of modules also precludes joint optimization of visual representation learning and temporal reasoning, leaving potential performance gains unrealized.

In contrast, the proposed Intent-Aware Structured (IAS) framework adopts a unified end-to-end architecture that eliminates error propagation entirely. By fine-tuning a single LLM with parameter-efficient LoRA, we achieve stable training without RL volatility. Using pre-extracted real intents as supervision avoids the sample inefficiency and overfitting risks inherent to RL-based intention learning. This holistic optimization yields robust generalization and competitive performance.

In summary, IAS offers a more robust and unified alternative to cascaded RL-based approaches. Nevertheless, several avenues for future work remain. First, integrating the process of structured video captioning into an end-to-end trainable framework could further enhance performance. Second, inferring high-quality intentions directly from observed video—without access to future ground truth could be an important direction for making intent-aware anticipation more practical. Finally, extending the IAS framework to handle even longer prediction horizons and complex hierarchical action structures could unlock new applications in more areas.